

Curriculum Skill Graph Pipeline: Steps 1 to 9

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Contents

0.1	Step 1: Data Audit and Preprocessing	2
0.1.1	Outputs of this Step	2
0.1.2	Fields to Fill After Concluding This Step	3
0.1.3	Implementation Note and Current Run	3
0.2	Step 2: Global Candidate Screening on Unique Textual Instances	4
0.2.1	Outputs of this Step	4
0.2.2	Fields to Fill After Concluding This Step	4
0.2.3	Implementation Note and Current Run	5
0.3	Step 3: Construction of the Manual Review Table	6
0.3.1	Outputs of this Step	6
0.3.2	Fields to Fill After Concluding This Step	6
0.3.3	Implementation Note and Current Run	6
0.4	Step 4: Harmonization Dataset Construction	8
0.4.1	Outputs of this Step	8
0.4.2	Fields to Fill After Concluding This Step	8
0.4.3	Implementation Note and Current Run	9
0.5	Step 5: Graph-Ready Node Construction Across Aggregation Levels	10
0.5.1	Outputs of this Step	10
0.5.2	Implementation Note and Current Run	10
0.6	Step 6: Full Weighted Graph Construction	12
0.6.1	Outputs of this Step	12
0.6.2	Implementation Note and Current Run	12
0.7	Step 7: Automatic Data-driven Selection of Representation and Graph Cutoff . .	14
0.7.1	Outputs of this Step	14
0.7.2	Implementation Note and Current Run	15
0.8	Step 8: Freeze and Report the Level-wise Selections	16
0.8.1	Outputs of this Step	16
0.8.2	Implementation Note and Current Run	16
0.9	Step 9: Descriptive Profiling and Overlap Analysis	17
0.9.1	Outputs of this Step	17
0.9.2	Implementation Note and Current Run	17

0.1 Step 1: Data Audit and Preprocessing

Step 1 converts the raw extracted JSON into a validated occurrence-level dataset that can support the later harmonization and graph-building stages. The step is intentionally non-destructive: it standardizes formatting, validates schema and vectors, expands list-like curricular provenance into occurrence rows, and records every repair or ambiguity in explicit audit files.

The expected raw fields are `skill_id`, `name`, `description`, `sections`, `programmes`, `year`, `edu_type`, `chunk_id`, `name_vector`, and `description_vector`. The implemented preprocessing verifies that these fields are present, parses the two embedding columns as numeric arrays, checks dimensional consistency, normalizes text and metadata conservatively, and expands the raw records to one row per curricular occurrence.

The key correction introduced in this pass concerns curricular provenance. Direct broadcasting of `sections`, `programmes`, and `year` is attempted first. When that broadcasting is not structurally sufficient, the script falls back to the `chunk_id`-encoded section-programme pairs together with a stable year lookup. These fallback repairs are not silently accepted: they are flagged with `provenance_resolution_confidence` and `requires_manual_validation`, and a sensitivity version of the cleaned dataset is exported with those ambiguous cases removed.

The output of Step 1 is therefore not only a cleaned occurrence table, but also an auditable record of what was standardized, what remained exact, what required conservative repair, and which cases still need manual confirmation before they are treated as fully trusted provenance.

0.1.1 Outputs of this Step

The corrected Step 1 produces the following files:

- `01_skills_occurrence_clean.json`
- `01_skills_occurrence_clean.csv`
- `01_skills_occurrence_clean_sensitivity_excluding_ambiguous.json`
- `01_skills_occurrence_clean_sensitivity_excluding_ambiguous.csv`
- `01_unique_textual_instances_vectors.csv`
- `01_data_audit_summary.csv`
- `01_schema_validation_summary.csv`
- `01_vector_integrity_report.csv`
- `01_vector_integrity_summary.csv`
- `01_duplicate_report.csv`
- `01_coverage_by_curricular_origin.csv`
- `01_metadata_standardization_log.csv`
- `01_preprocessing_decision_log.csv`
- `01_provenance_resolution_audit.csv`
- `01_step1_deviations_notes.txt`

The cleaned occurrence dataset now includes the fields `text_instance_id`, `provenance_resolution_confidence`, and `requires_manual_validation`. The file `01_unique_textual_instances_vectors.csv` is the corrected screening base used in Step 2.

0.1.2 Fields to Fill After Concluding This Step

No extra analyst input is required to complete the computational part of Step 1. The only remaining manual task is to inspect the flagged provenance repairs recorded in:

- `01_preprocessing_decision_log.csv`
- `01_provenance_resolution_audit.csv`

These files identify the raw records whose occurrence expansion depends on fallback provenance reconstruction and should therefore be checked before the fully trusted node universe is frozen.

0.1.3 Implementation Note and Current Run

The current corrected run produced:

- 895 raw records;
- 1,038 occurrence rows after preprocessing;
- 26 raw records with list-length mismatch that required chunk-based provenance repair;
- 88 occurrence rows flagged with `requires_manual_validation = true`;
- 950 rows in the sensitivity dataset that excludes those ambiguous cases;
- 895 unique textual instances in `01_unique_textual_instances_vectors.csv`;
- 0 invalid `name_vector` rows and 0 invalid `description_vector` rows;
- 0 exact duplicate occurrence groups.

The main documented implementation choices are:

- direct metadata broadcasting is preferred, with chunk-based reconstruction used only as a fallback;
- duplicate detection is exact-only at the occurrence level;
- normalization is conservative and does not rewrite the semantic content of the extracted skills;
- array-valued fields are serialized as JSON strings in CSV exports for traceability.

0.2 Step 2: Global Candidate Screening on Unique Textual Instances

After Step 1 has produced a validated occurrence-level base, the next task is to screen for potentially equivalent skill labels at the global corpus level. This step is not a graph-construction step. Its purpose is to build a candidate table for later review and harmonization, not to choose the final sparse graph cutoff.

The corrected Step 2 uses `01_unique_textual_instances_vectors.csv` as its direct input. This file contains one row per unique textual instance rather than one row per unique skill name. The distinction matters methodologically: exact-name matches are no longer collapsed in advance, so the workflow does not hard-code the assumption that identical names must always correspond to the same harmonized skill. If exact-name duplicates exist, they are placed in a dedicated review stratum rather than treated as automatic matches.

For each unordered pair of textual instances, Step 2 computes cosine similarity under two fixed representations:

- `name_only`, using the normalized `name_vector`;
- `name_description`, using the normalized concatenation of `name_vector` and `description_vector`.

The result is a global screening table that records both similarity values. A separate screening-cutoff file records the current provisional thresholds that define which pairs move forward to Step 3 review. Those thresholds are explicitly described as review-screening thresholds only. Final graph sparsification is deferred to Steps 7 and 8.

0.2.1 Outputs of this Step

The corrected Step 2 produces:

- `02_candidate_pairwise_similarity.csv`
- `02_candidate_screening_summary.csv`
- `02_screening_cutoff_selection.csv`
- `02_exact_name_candidate_stratum.csv`
- `02_screening_calibration_template.csv`

The pairwise table is the main output. Each row represents one unordered pair of textual instances and includes the two similarity scores plus the identifiers and labels needed for downstream review. The exact-name stratum file is retained even if empty so that the handling of exact-name cases remains explicit.

0.2.2 Fields to Fill After Concluding This Step

The only file that may require analyst input at this stage is:

- `02_screening_cutoff_selection.csv`

In the corrected workflow, this file stores provisional screening thresholds and their status. The intended next improvement is pilot calibration on a labeled sample, because the methodological objective here is high review recall at manageable manual-review load, not direct graph optimization.

0.2.3 Implementation Note and Current Run

The current corrected run used `01_unique_textual_instances_vectors.csv` as input and produced:

- 895 textual instances;
- 400,065 unordered candidate pairs;
- 0 exact-name candidate-stratum pairs in the present corpus;
- 48 rows in the calibration template.

The current screening summaries report:

- `name_only`: p95 = 0.315562
- `name_description`: p95 = 0.305355

The currently recorded screening cutoffs are:

- `screening_cutoff_name_only` = 0.70622
- `screening_cutoff_name_description` = 0.701394
- `cutoff_source` = `analyst_fixed_provisional`
- `cutoff_status` = `pending_pilot_calibration`

Pairs are sent to Step 3 if they pass at least one of these two screening cutoffs. This union rule is used only to form the review list and does not constitute a representation choice for graph construction.

0.3 Step 3: Construction of the Manual Review Table

Step 3 converts the Step 2 candidate-screening output into a manual-review table. This step is intentionally a review stage, not a merge stage. It does not assign canonical names, create harmonized groups, or collapse the node universe. Its sole purpose is to identify which candidate pairs should be examined before any equivalence decision is propagated downstream.

The direct inputs are `02_candidate_pairwise_similarity.csv`, `02_screening_cutoff_selection.csv`, and `02_exact_name_candidate_stratum.csv`. Candidate pairs are retained if they satisfy at least one of the provisional screening cutoffs. Exact-name pairs, if any exist, remain a separate review stratum rather than an automatic merge rule. The review template also carries forward whether either textual instance was flagged in Step 1 as requiring manual provenance validation.

The corrected design restores the three allowed review states:

- `equivalent`
- `not_equivalent`
- `uncertain`

The columns `review_decision` and `review_notes` are left blank by default. No automatic equivalence-by-cutoff rule is applied.

0.3.1 Outputs of this Step

The corrected Step 3 produces:

- `03_review_pairs_textual_instances.csv`
- `03_review_summary.csv`
- `03_review_template_textual_instances.csv`

The review template is the file to be completed by the analyst before true harmonization can begin. It retains the similarity evidence from Step 2 together with the empty review fields and any provenance warning indicators inherited from Step 1.

0.3.2 Fields to Fill After Concluding This Step

The following columns in `03_review_template_textual_instances.csv` are meant to be completed after inspection:

- `review_decision`
- `review_notes`

Until those fields are filled, Step 4 should remain in a no-merge baseline.

0.3.3 Implementation Note and Current Run

The current corrected run retained:

- 160 total review pairs;
- 144 pairs passing the `name_only` screening cutoff;

- 57 pairs passing the `name_description` screening cutoff;
- 41 pairs passing both screening cutoffs;
- 16 review pairs involving at least one text instance flagged for manual provenance validation;
- 0 exact-name review-stratum pairs.

Most importantly, the current review file is unfilled:

- `review_decisions_pre_filled = false`
- `allowed_review_states = equivalent;not_equivalent;uncertain`

This means Step 3 is now aligned with the intended methodology. The workflow pauses here for actual judgment rather than converting screening cutoffs directly into harmonization decisions.

0.4 Step 4: Harmonization Dataset Construction

Step 4 transforms reviewed equivalence decisions into a harmonization map that can be propagated back to the full occurrence-level dataset. In the corrected workflow, this step is deliberately conservative: only pairs explicitly marked **equivalent** in Step 3 are allowed to create harmonized groups. Pairs marked **not_equivalent** or **uncertain**, as well as pairs with blank review fields, do not create merges.

The grouping rule is connected-component closure over the reviewed positive-pair graph. Because chain merges can over-collapse the node universe when the review edges are noisy, Step 4 now exports an explicit component audit so that multi-name or multi-member groups can be checked before the harmonization layer is treated as frozen. Representative harmonized names are selected transparently by the rule “most frequent label, then lexicographic tie-break,” with optional manual overrides recorded separately.

If Step 3 has not yet been reviewed, Step 4 remains valid but yields a no-merge baseline. This is the current pipeline state.

0.4.1 Outputs of this Step

The corrected Step 4 produces:

- 04_textual_instances_harmonization_mapping.csv
- 04_textual_instances_harmonized.csv
- 04_skills_harmonized.json
- 04_skills_harmonized.csv
- 04_unique_skills_harmonization_mapping.csv
- 04_unique_skills_harmonized.csv
- 04_harmonization_summary.csv
- 04_harmonized_groups_summary.csv
- 04_harmonized_name_selection.csv
- 04_component_merge_audit.csv

The textual-instance mapping is the main harmonization file for downstream steps. The unique-skill summary files are retained as derived convenience outputs, but the harmonization logic itself is defined at the textual-instance level.

0.4.2 Fields to Fill After Concluding This Step

No new fields are meant to be filled inside Step 4 itself. The substantive prerequisite is still upstream:

- complete the Step 3 review fields **review_decision** and **review_notes** in **03_review_template_textual_instances.csv**

After those decisions are entered, Step 4 should be rerun so that the harmonized node universe reflects reviewed equivalence rather than the current no-merge baseline.

0.4.3 Implementation Note and Current Run

The current corrected run reports:

- reviewed_pairs_equivalent = 0
- reviewed_pairs_not_equivalent = 0
- reviewed_pairs_uncertain = 0
- review_pairs_without_decision = 160
- total_text_instances_before_harmonization = 895
- total_harmonized_groups = 895
- multi_text_instance_harmonized_groups = 0
- text_instances_name_changed = 0
- full_skill_records_name_changed = 0
- harmonization_mode = no_merge_baseline

The current Step 4 outputs are therefore methodologically valid, but they still describe the pre-review baseline. Any actual harmonization requires reviewed **equivalent** decisions to be entered in Step 3 and then propagated through a rerun of Steps 4 to 9.

0.5 Step 5: Graph-Ready Node Construction Across Aggregation Levels

Step 5 freezes the analytical units and constructs the graph-ready node tables on which all later weighted and sparse graphs are built. The corrected workflow makes every aggregation level explicit. The node universe is therefore defined for:

- the pooled all-tracks unit;
- track units;
- section units;
- programme units;
- programme-year units.

Within each unit, all full-sample occurrence rows carrying the same `harmonized_skill_id` are collapsed into one graph node. The node embeddings are aggregated consistently across representations: normalized `name_vector` values are averaged and renormalized to form `node_name_vector`, while normalized concatenated `name + description` vectors are averaged and renormalized to form `node_name_description_vector`. This aggregation rule is held fixed across all later representation comparisons.

The node tables retain provenance counts and memberships so that downstream overlap and recurrence analyses can still be traced back to the underlying curricular structure.

0.5.1 Outputs of this Step

The corrected Step 5 produces:

- `05_analysis_units_definition.csv`
- `05_unit_and_aggregation_decisions.csv`
- `05_graph_ready_nodes_pooled.csv`
- `05_graph_ready_nodes_tracks.csv`
- `05_graph_ready_nodes_sections.csv`
- `05_graph_ready_nodes_programmes.csv`
- `05_graph_ready_nodes_programme_years.csv`
- `05_node_aggregation_summary.csv`
- `05_harmonized_node_provenance.csv`

0.5.2 Implementation Note and Current Run

The current corrected run defines:

- 1 pooled unit;
- 2 track units;
- 13 section units;
- 14 programme units;

- 33 programme-year units.

The current node totals are:

- pooled: 895 graph nodes from 1,038 source records;
- tracks: 895 graph nodes across the two track tables;
- sections: 1,036 graph nodes across all section tables;
- programmes: 969 graph nodes across all programme tables;
- programme-years: 1,037 graph nodes across all programme-year tables.

Because Step 4 is currently a no-merge baseline, these node tables collapse repeated occurrences within units but do not yet reflect cross-label reviewed harmonization merges.

0.6 Step 6: Full Weighted Graph Construction

Step 6 converts each Step 5 node table into a full weighted graph under both semantic representations. This step is mechanical but important: it freezes the common weighted candidate graph family from which later sparsification and selection are derived.

For every analytical unit, Step 6 computes cosine similarity between all unordered node pairs under:

- `name_only`, using `node_name_vector`;
- `name_description`, using `node_name_description_vector`.

The corrected workflow now makes this graph family explicit for all five aggregation levels: pooled, track, section, programme, and programme-year. The weighted edges are stored in long format together with summary metadata so that later steps can reuse the same frozen weighted universe regardless of the eventual sparsification decision.

0.6.1 Outputs of this Step

The corrected Step 6 produces:

- `06_similarity_matrices_pooled.csv`
- `06_similarity_matrices_tracks.csv`
- `06_similarity_matrices_sections.csv`
- `06_similarity_matrices_programmes.csv`
- `06_similarity_matrices_programme_years.csv`
- `06_weighted_edges_pooled.csv`
- `06_weighted_edges_tracks.csv`
- `06_weighted_edges_sections.csv`
- `06_weighted_edges_programmes.csv`
- `06_weighted_edges_programme_years.csv`
- `06_weighted_graph_summary.csv`
- `06_weighted_graph_metadata.csv`
- `06_graph_storage_decisions.csv`

0.6.2 Implementation Note and Current Run

The current corrected run built weighted graphs for:

- 1 pooled unit;
- 2 track units;
- 13 section units;
- 14 programme units;

- 33 programme-year units.

Across the two semantic representations, this yields 126 weighted graphs in total. The pooled weighted graphs are built on 895 nodes and 400,065 unordered edges per representation. All later sparsification decisions in Step 7 operate on this frozen weighted family.

0.7 Step 7: Automatic Data-driven Selection of Representation and Graph Cutoff

Step 7 now follows the data-driven design in `data_driven.tex`. The selection problem is no longer framed as a topology heuristic based on bootstrap edge Jaccard, susceptibility, or a pooled fallback rule. Instead, for each aggregation level separately, the workflow jointly selects:

- the semantic representation; and
- the graph cutoff.

The aggregation levels are `pooled`, `track`, `section`, `programme`, and `programme_year`. The candidate semantic representations are `name_only` and `concat` (implemented from the Step 6 `name_description` weights). Selection is performed on the frozen no-merge node universe inherited from Steps 4 and 5.

For each level and representation, the candidate threshold set is the set of distinct observed Step 6 edge weights at that level. The selection objective is the cross-validated skill score described in `data_driven.tex`: holdout edge retention is compared against a holdout base-rate benchmark using log-loss, and the best representation-threshold pair is chosen separately for each level.

The practical implementation keeps the Step 6 edge weights fixed and resamples at the occurrence level through node-presence variation. Bootstrap edge-retention probabilities are computed analytically from train-fold occurrence counts rather than estimated by Monte Carlo simulation. This is exact for units in which each node is supported by a single occurrence and an approximation for the small subset of units with collapsed repeated occurrences.

If the best skill at a level is strictly greater than zero, the selected output for that level is a thresholded sparse graph. If the best skill does not exceed zero, the automatic decision is to retain the full weighted graph for that level. Step 7 therefore returns a final selected graph family, not only a sparse-only family.

0.7.1 Outputs of this Step

The current Step 7 produces:

- `07_levelwise_model_selection_surface.csv`
- `07_levelwise_selected_models.csv`
- `07_selected_graphs_pooled.csv`
- `07_selected_graphs_tracks.csv`
- `07_selected_graphs_sections.csv`
- `07_selected_graphs_programmes.csv`
- `07_selected_graphs_programme_years.csv`
- `07_edge_selection_probabilities_pooled.csv`
- `07_edge_selection_probabilities_tracks.csv`
- `07_edge_selection_probabilities_sections.csv`
- `07_edge_selection_probabilities_programmes.csv`
- `07_edge_selection_probabilities_programme_years.csv`
- `07_selected_graph_summary.csv`
- `07_selection_decision_log.csv`

0.7.2 Implementation Note and Current Run

The current run selected the following level-wise models:

- pooled: `name_only` at threshold 0.814399, with `sparsification_supported` = `TRUE`;
- track: `name_only` at threshold 0.819634, with `sparsification_supported` = `FALSE`;
- section: `name_only` at threshold 0.831524, with `sparsification_supported` = `FALSE`;
- programme: `concat` at threshold 0.879671, with `sparsification_supported` = `FALSE`;
- programme-year: `concat` at threshold 0.790938, with `sparsification_supported` = `FALSE`.

This means that, in the present run, only the pooled graph is exported as a thresholded sparse graph. All other levels revert to their full weighted graphs because no thresholded model outperformed the holdout base-rate benchmark. The selected graph summary therefore contains:

- 1 pooled thresholded graph;
- 2 track full weighted graphs;
- 13 section full weighted graphs;
- 14 programme full weighted graphs;
- 33 programme-year full weighted graphs.

The pooled selected graph retains only 4 edges in the current run. The later descriptive atlas and final export package therefore inherit a mixed graph family: one thresholded pooled graph and full weighted graphs at the remaining aggregation levels.

0.8 Step 8: Freeze and Report the Level-wise Selections

Under the corrected workflow, the substantive representation-and-cutoff choice is already complete in Step 7. Step 8 therefore acts only as a freeze-and-report layer. Its purpose is to copy the selected models into a stable downstream record, document that the graph family is indexed by aggregation level, and make that frozen selection explicit for later atlas construction and export. The direct input is `07_levelwise_selected_models.csv`. Unlike earlier versions of the workflow, Step 8 does not run a second composite selector. The full candidate surface remains in Step 7, and Step 8 simply records the winning row for each level.

0.8.1 Outputs of this Step

The current Step 8 produces:

- `08_levelwise_alternatives.csv`
- `08_primary_representation_selection.csv`
- `08_levelwise_selection_report.csv`
- `08_selection_freeze_log.csv`

0.8.2 Implementation Note and Current Run

The current run freezes five level-wise selections:

- pooled: `name_only`, threshold 0.814399, `thresholded_sparse`;
- track: `name_only`, threshold 0.819634, `full_weighted_graph`;
- section: `name_only`, threshold 0.831524, `full_weighted_graph`;
- programme: `concat`, threshold 0.879671, `full_weighted_graph`;
- programme-year: `concat`, threshold 0.790938, `full_weighted_graph`.

The freeze log records that:

- Step 8 is `freeze_and_report_only`;
- the selection scope is level-wise rather than global;
- the selection source is `07_levelwise_selected_models.csv`;
- the full candidate surface remains in Step 7 rather than being duplicated in Step 8.

This means the downstream descriptive atlas and final export package should be interpreted as a mixed level-wise graph family, not as one universal representation-cutoff rule applied everywhere.

0.9 Step 9: Descriptive Profiling and Overlap Analysis

Step 9 builds the descriptive atlas on the frozen graph family produced by Steps 7 and 8. Because the current data-driven selector may keep a thresholded sparse graph at one level and a full weighted graph at another, the atlas now carries level-specific graph-mode metadata in addition to level-specific representation and cutoff metadata.

The step contains four descriptive blocks:

- graph profiles for every analytical unit;
- within-programme skill recurrence across years;
- programme-year progression overlap;
- cross-unit overlap relative to tracks, sections, programmes, programme-years, and the pooled universe.

The current implementation reads the final selected graph files from Step 7, not the retired sparse-only files. When a level has `selected_graph_mode = full_weighted_graph`, the atlas still reports graph sizes and node overlaps, but edge-overlap diagnostics are intentionally left blank because they are not substantively informative for complete weighted graphs.

0.9.1 Outputs of this Step

The current Step 9 produces:

- `09_graph_profile_table.csv`
- `09_programme_skill_recurrence_summary.csv`
- `09_programme_skill_recurrence_detail.csv`
- `09_programme_year_adjacent_progression_overlap.csv`
- `09_programme_year_progression_overlap_all_pairs.csv`
- `09_overlap_matrix_tracks.csv`
- `09_overlap_matrix_sections.csv`
- `09_overlap_matrix_programmes.csv`
- `09_overlap_matrix_programme_years.csv`
- `09_overlap_to_pooled.csv`
- `09_descriptive_atlas_summary.csv`

0.9.2 Implementation Note and Current Run

The current atlas summary reports:

- 5 frozen selection rows inherited from Step 8;
- 63 graph-profile rows;
- 14 programme recurrence summaries;
- 969 programme recurrence detail rows;

- 19 adjacent-year progression rows;
- 29 all-pairs progression rows;
- 1 track-overlap row;
- 78 section-overlap rows;
- 91 programme-overlap rows;
- 528 programme-year-overlap rows;
- 62 pooled-comparison rows.

In the present run, Step 9 inherits a mixed graph family:

- pooled: thresholded sparse graph under `name_only`;
- track: full weighted graph under `name_only`;
- section: full weighted graph under `name_only`;
- programme: full weighted graph under `concat`;
- programme-year: full weighted graph under `concat`.

This atlas therefore remains tied to the Step 4 no-merge baseline, but it no longer assumes that every level is sparsified. The current descriptive package should be read as the atlas for the data-driven selected graph family, not as the atlas for a universal sparse-graph rule.